

Take the integral:

$$\int \frac{1}{4 \cos^4(x) + 4 \cos^2(x) + 1} dx$$

Multiply numerator and denominator of $\frac{1}{4 \cos^4(x) + 4 \cos^2(x) + 1}$ by $\sec^4(x)$:

$$= \int \frac{\sec^4(x)}{\sec^4(x) + 4 \sec^2(x) + 4} dx$$

Prepare to substitute $u = \tan(x)$. Rewrite $\frac{\sec^4(x)}{\sec^4(x) + 4 \sec^2(x) + 4}$ using $\sec^2(x) = \tan^2(x) + 1$:

$$= \int \frac{(1 + \tan^2(x)) \sec^2(x)}{(3 + \tan^2(x))^2} dx$$

For the integrand $\frac{(1 + \tan^2(x)) \sec^2(x)}{(3 + \tan^2(x))^2}$, substitute $u = \tan(x)$ and $du = \sec^2(x) dx$:

$$= \int \frac{u^2 + 1}{(u^2 + 3)^2} du$$

For the integrand $\frac{u^2 + 1}{(u^2 + 3)^2}$, use partial fractions:

$$= \int \left(\frac{1}{u^2 + 3} - \frac{2}{(u^2 + 3)^2} \right) du$$

Integrate the sum term by term and factor out constants:

$$= \int \frac{1}{u^2 + 3} du - 2 \int \frac{1}{(u^2 + 3)^2} du$$

Factor 3 from the denominator:

$$= \int \frac{1}{3 \left(\frac{u^2}{3} + 1 \right)} du - 2 \int \frac{1}{(u^2 + 3)^2} du$$

Factor out constants:

$$= \frac{1}{3} \int \frac{1}{\frac{u^2}{3} + 1} du - 2 \int \frac{1}{(u^2 + 3)^2} du$$

For the integrand $\frac{1}{\frac{u^2}{3} + 1}$, substitute $s = \frac{u}{\sqrt{3}}$ and $ds = \frac{1}{\sqrt{3}} du$:

$$= \frac{1}{\sqrt{3}} \int \frac{1}{s^2 + 1} ds - 2 \int \frac{1}{(u^2 + 3)^2} du$$

The integral of $\frac{1}{s^2 + 1}$ is $\tan^{-1}(s)$:

$$= \frac{\tan^{-1}(s)}{\sqrt{3}} - 2 \int \frac{1}{(u^2 + 3)^2} du$$

For the integrand $\frac{1}{(u^2+3)^2}$, substitute $u = \sqrt{3} \tan(p)$ and $du = \sqrt{3} \sec^2(p) dp$. Then $(u^2 + 3)^2 = (3 \tan^2(p) + 3)^2 = 9 \sec^4(p)$ and $p = \tan^{-1}\left(\frac{u}{\sqrt{3}}\right)$:

$$= \frac{\tan^{-1}(s)}{\sqrt{3}} + -2 \sqrt{3} \int \frac{\cos^2(p)}{9} dp$$

Factor out constants:

$$= \frac{\tan^{-1}(s)}{\sqrt{3}} - \frac{2}{3 \sqrt{3}} \int \cos^2(p) dp$$

Write $\cos^2(p)$ as $\frac{1}{2} \cos(2p) + \frac{1}{2}$:

$$= \frac{\tan^{-1}(s)}{\sqrt{3}} - \frac{2}{3 \sqrt{3}} \int \left(\frac{1}{2} \cos(2p) + \frac{1}{2} \right) dp$$

Integrate the sum term by term and factor out constants:

$$= \frac{\tan^{-1}(s)}{\sqrt{3}} - \frac{1}{3 \sqrt{3}} \int \cos(2p) dp - \frac{1}{3 \sqrt{3}} \int 1 dp$$

For the integrand $\cos(2p)$, substitute $w = 2p$ and $dw = 2 dp$:

$$= \frac{\tan^{-1}(s)}{\sqrt{3}} - \frac{1}{6 \sqrt{3}} \int \cos(w) dw - \frac{1}{3 \sqrt{3}} \int 1 dp$$

The integral of $\cos(w)$ is $\sin(w)$:

$$= \frac{\tan^{-1}(s)}{\sqrt{3}} - \frac{\sin(w)}{6 \sqrt{3}} - \frac{1}{3 \sqrt{3}} \int 1 dp$$

The integral of 1 is p :

$$= -\frac{p}{3 \sqrt{3}} + \frac{\tan^{-1}(s)}{\sqrt{3}} - \frac{\sin(w)}{6 \sqrt{3}} + \text{constant}$$

Substitute back for $w = 2p$:

$$= -\frac{p}{3 \sqrt{3}} - \frac{\sin(2p)}{6 \sqrt{3}} + \frac{\tan^{-1}(s)}{\sqrt{3}} + \text{constant}$$

Substitute back for $p = \tan^{-1}\left(\frac{u}{\sqrt{3}}\right)$:

$$= \frac{1}{9} \left(3 \sqrt{3} \tan^{-1}(s) - \frac{3u}{u^2 + 3} - \sqrt{3} \tan^{-1}\left(\frac{u}{\sqrt{3}}\right) \right) + \text{constant}$$

Substitute back for $s = \frac{u}{\sqrt{3}}$:

$$= \frac{1}{9} \left(2 \sqrt{3} \tan^{-1}\left(\frac{u}{\sqrt{3}}\right) - \frac{3u}{u^2 + 3} \right) + \text{constant}$$

Substitute back for $u = \tan(x)$:

$$= \frac{1}{9} \left(2 \sqrt{3} \tan^{-1}\left(\frac{\tan(x)}{\sqrt{3}}\right) - \frac{3 \tan(x)}{\tan^2(x) + 3} \right) + \text{constant}$$

Which is equal to:

Answer:

$$= \frac{2 \tan^{-1}\left(\frac{\tan(x)}{\sqrt{3}}\right)}{3 \sqrt{3}} - \frac{\sin(2x)}{6 (\cos(2x) + 2)} + \text{constant}$$